and young adults. Meningococci are strict human pathogens. Human nasopharynx is the reservoir of *N.meningitidis.* The pathogenesis is dicussed in the flowchart 7.2.

Source of infection – Airborne droplets

Route of entry – Nasopharynx

Site of infection – Meninges

Incubation period – 3 days

7.5.4 Laboratory Diagnosis

Specimens: CSF, blood, nasopharyngeal scrapings from petechiae lesions are the specimens collected from pyogenic meningitis patients.

Direct Microscopy: CSF is centrifuged, and smear is prepared from the deposit for gram staining. Meningococci are Gram negative diplococci, present mainly inside polymorphs and many pus cells are also seen.

Culture: The centrifuged deposit of CSF is inoculated on chocolate agar. The plate is incubated at 36°C under 5–10% CO_2 for 18–24 hours. After incubation period, meningococcusis identified by gram staining, colony morphology and biochemical reactions. *N. meningitides* is catalase and oxidase positive (Figure 7.8).

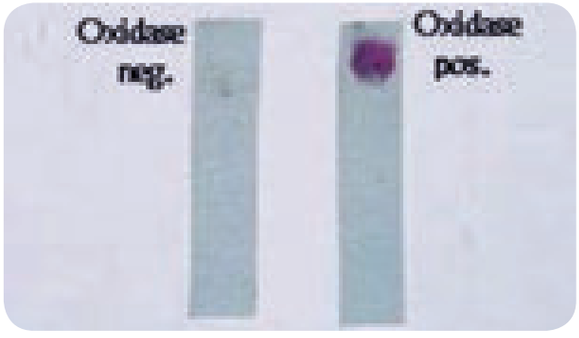


Figure 7.8: Oxidase test

The meningococci gain entry in the nasopharynx and attaches to the epithelial cells with pili. They are engulfed by epithelial cells of mucosa and penetratesintonearby blood vessels, thereby damaging the epithelium and causes pharyngitis.

↓

Cocci spread from the nasopharynx to meninge by travelling along the perineural sheath of the olfactory nerve, through the cribriform plate to the subarachnoid space or through blood stream.

↓

Pathogen entering the blood vessels rapidly permeates the meninges and produce meningitis (most complication in children). It is marked by the following clinical manifestations, which are fever, sore throat, headache, stiff neck, and vomiting convulsions (fits).

↓

The pathogen sheds endotoxin into the generalized circulation, which damages the blood vessels and leads to vascular collapse, hemorrhage, petechiae lesion (a small red or purple spot caused by bleeding into the skin).

↓

A few develop sudden meningococcemia **(water house – friderichsen syndrome)** characterized by shock, disseminated intravascular coagulation and multisystem failure.

↓

It has a violet onset, with fever, chills, shock and coma. Generalized intravascular clotting, cardiac failure, damage to adrenal glands and death occurs within a few hours.

Flowchart 7.2: Pathogenesis of *Neisseria Meningitides*

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